

Name: Bell, Samuel C.

Date: 07/15/2026

Scenario:

We wish to create a database for a company that runs training courses. For this, we must store data about the trainees and the instructors. For each course participant (about 500), identified by a code, we want to store the social security number, surname, age, sex, place of birth, employer's name, address and telephone number, previous employers (and period employed), the courses attended (there are about 200 courses) and the final assessment of each course. We need also to represent the seminars that each participant is attending at present and, for each day, the places and times the classes are held. Each course has a code and a title and any course can be given any number of times. Each time a particular course is given, we will call it an 'edition' of the course. For each edition, we represent the start date, the end date, and the number of participants. If a trainee is self-employed professional, we need to know his or her area of expertise, and, if appropriate, his or her title. For somebody who works for a company, we store the level and position held. For each instructor (about 300), we will show the surname, age, place of birth, the edition of the course taught, those taught in the past and the courses that the tutor is qualified to teach. All the instructors' telephone numbers are also stored. An instructor can be permanently employed by the training company or can be freelance.

1. Analyze the scenario given and identify the following:

- Organization goals – what are the tasks that are needed to accomplish?

Keep accurate records of all the trainees (personal info, employment, SSN), keep accurate records of all instructors (qualifications, contact info), offer and manage courses and their editions, schedule classes without time or place conflicts, assign qualified instructors to each edition, and record and track each trainee's final assessment per course.

- Business functions – what are the different processes?

The first is Trainee Enrollment/Registration, second is Employment Record Management (current and previous employers), third is Course and Edition Management, fourth is Class Scheduling, fifth is Instructor Assignment and Qualification Tracking, and the last one, Assessment Recording.

- Entities – what are the different objects included in the database system?

For Trainee, it's the code, SSN, surname, age, sex, place of birth, address, phone, employer info, self-employed details or employee details.

For Employer, it has to be the company name the trainee work/worked for.

For Course, it's the code and title.

For Edition, a specific run of a course like the starting date, end date, and number of participants.

For Class Schedule, the day, place, and time of each session within an edition.

And for Assessment, the trainee's final grade or result per course.

2. Create a Planning Matrix using the Function-to- data entity matrix type

Business Functions \ Data Entity Types	Customer	Product	Raw Material	Order	Work Center	Work Order	Invoice	Equipment	Employee
Business Planning	X	X						X	X
Product Development		X	X		X			X	
Materials Management		X	X	X	X	X		X	
Order Fulfillment	X	X	X	X	X	X	X	X	X
Order Shipment	X	X		X	X		X		X
Sales Summarization	X	X		X			X		X
Production Operations		X	X	X	X	X		X	X
Finance and Accounting	X	X	X	X	X		X	X	X

X = data entity (column) is used within business function (row)

	Trainee	Employer	Course	Edition	Class Schedule	Instructor	Assessment
Trainee Enrollment	X	X					
Employment Record Management	X	X					
Course & Edition Management			X	X			
Class Scheduling				X	X		
Instructor Assignment & Qualification			X	X		X	
Assessment Recording	X		X	X			X

X = data entity (column) is used within business function (row)